

**Economics After Cognitive Scarcity:
The Permanent-Remainder Assumption and What Would
Have to Be True for It to Hold**

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Abstract

Many arguments about the effect of artificial intelligence on work assume that tasks machines cannot yet perform form a lasting human domain. We call this the permanent-remainder assumption and examine what it requires. Under a counterfactual in which every attainable cognitive service is free and universally available, we solve three benchmark models. In a two-good equilibrium with identical preferences and identical free cognition, the ratio of consumption between two groups is 81 to 1, 9 to 1 or 1 to 1 depending only on ownership and pooling, so equal access to intelligence does not by itself narrow the distribution of consumption. In a task model, the set of tasks held by people can be nonempty at every date while their share of tasks converges to $\delta/(a + \delta)$, where δ is the rate of task turnover and a the rate at which capability overtakes a new task. This limit does not depend on the rate at which new tasks appear, and it is positive only if some tasks require something other than reasoning. In a model of fallback after automated incident response, a human responder is the best fallback in 84.85 percent of the plane of machine capability and failure independence, and beyond 43.2 percent automation of routine incidents an out-of-practice responder performs worse than simple containment. The models are benchmarks with no prices, investment or empirical calibration.

Introduction

A planner in 1900 asked to prepare for a much larger New York could have produced a correct estimate of how much horse manure the city would have to remove. The estimate would have been accurate and irrelevant, because it treated the horse as a fixed feature of urban transport. Current arguments about artificial intelligence and work make a similar move. They can grant machines very high capability and still describe the future as the present with different staffing: machines handle routine cases and people handle exceptions, machines produce answers and people check them, old occupations disappear and new ones appear. Each version keeps a fixed human role and relocates it within an otherwise unchanged arrangement.

The Lucas critique warned against treating estimated behavioural relations as invariant when a policy changes the environment in which they were estimated (Lucas 1976). The present case differs, since the change is in technological capability and organisation, but it shares the structure: a relation can hold within one arrangement and say nothing about the transition to another.

The possibility of machine-dominated production has been discussed for a long time. Simon predicted in 1960 that machines would within twenty years be capable of any work a person could do (Simon 1960). Marschak treated the cost of thinking as an economic input (Marschak 1968), and Nilsson analysed artificial intelligence, employment and income as a single problem (Nilsson 1985). Romer's growth theory rests on the distinction between a non-rival design and a person's rival capacity to execute it (Romer 1990), and Meade linked machine production to ownership and social dividends (Meade 1993). Recent work models the automation of production and of research directly (Trammell and Korinek 2026; Growiec et al. 2023). We ask which inferences survive when

the premise of free cognition is applied consistently, including to the activities used to preserve the existing arrangement.

We state the counterfactual precisely. Every attainable cognitive service, at every quality, is free to develop, copy and run, and every person and institution has unrestricted access. This includes reasoning about science, engineering, contracts, management, verification, persuasion, organisational design and public policy, and it includes finding errors in plans and designing better ways to coordinate. It excludes four things: omniscience, because inference cannot produce observations that were never made; free embodiment, because a perfect controller is not a physical machine; agreement, because people who understand each other fully can still want incompatible things; and authority, because knowing how to do something does not confer permission to do it. Under this premise, a lasting human role can arise only from one of these four exclusions.

Distribution under identical intelligence

Consider an economy with 100 people, a general consumption good and a housing service, both in fixed net supply because their physical production is already automated. Preferences are identical, $\log c + \eta \log h$, and free cognition enters every utility function as the same additive constant, so it does not affect how the two scarce goods are divided. Person i owns a share θ_i of every productive asset, and a fraction τ of asset income is pooled and shared equally. In the competitive equilibrium each person's shares of consumption and housing are both $s_i = (1 - \tau)\theta_i + \tau/N$, and the relative price of housing is $\eta Q/H$.

Let ten people own 90 percent of the assets and the other ninety own the rest, with both goods in supply 100. Without pooling, each of the ten consumes 9.000 and each of the ninety consumes 0.111, a ratio of 81 to 1. Pooling half of asset income gives 5.000 against 0.556, a ratio of 9 to 1, and pooling all of it gives 1.000 each. Intelligence, preferences, technology and total supplies are identical in all three cases.

The construction is standard, and its purpose is to show what equal intelligence does not determine. For any strictly positive vector of consumption shares summing to one, some ownership vector supports it; this was checked against 200 randomly drawn target distributions with exact agreement. Universal access to cognition is therefore consistent with every distribution of consumption. A claim that free intelligence will equalise outcomes, or that it will concentrate them, requires an argument about ownership.

Multiplying the supply of the consumption good tenfold, with nothing else changed, raises everyone's consumption of it tenfold. Housing per person is unchanged, at 9.000 and 0.111, and the relative price of housing rises from 1.0 to 10.0. Abundance of goods in general does not imply greater access to a good whose supply is fixed.

The permanent-remainder assumption

The permanent-remainder assumption holds that whatever machines cannot yet do forms a lasting human domain. It appears in several forms, such as the division of routine and exceptional work,

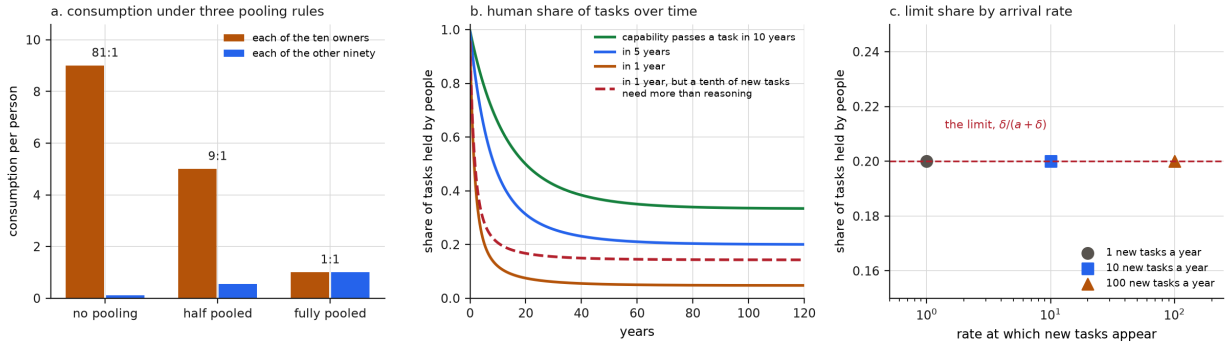


Figure 1: Allocation and the task frontier. (a) Consumption per person in each group under three pooling rules, with intelligence, preferences and physical supplies identical. (b) The share of tasks held by people over time, for three rates at which capability overtakes a new task, and for the case in which a tenth of new tasks require something other than reasoning. (c) The same limit for arrival rates spanning a hundredfold range.

of answering and verifying, and of disappearing and newly created jobs. These forms share an inference from a statement about every date to a statement about an enduring set of tasks, and the inference is invalid: that at every date some task favours people does not imply that some task favours people at every date.

To make this concrete, give each task a capability requirement and let machine capability rise over time, so that a task is held by people while capability is below its requirement. A task created above the capability frontier is overtaken at rate a , every task becomes obsolete at rate δ , and new tasks appear at rate ν . The share of tasks held by people then converges to

$$\frac{\delta}{a + \delta},$$

and the simulation reproduces this limit to within 5.50×10^{-5} across a grid of turnover and overtaking rates.

The set of tasks held by people is never empty. In every run the stock of human-held tasks is strictly positive at every date, because new tasks keep arriving above the frontier. This is compatible with a human share of tasks of 0.048 when capability overtakes a new task within a year and turnover is 5 percent a year. The informal argument treats a nonempty frontier as evidence of a substantial share, and the model shows that the two are independent.

The arrival rate does not affect the limit. Varying ν from 1 to 100 new tasks a year changes the limiting share by 0.000000. New and old tasks accumulate in the same proportion, so the share held by people is set by turnover and overtaking alone. Arguments that rely on the volume of new work therefore do not bear on the share of work people retain.

A positive limit requires tasks with a requirement other than reasoning. If a share of new tasks has a requirement that capability never meets, which under the premise means physical presence, an

unavailable observation, consent or legitimate authority, the limiting share rises. With a tenth of new tasks of this kind it rises from 0.048 to 0.143. A lasting human role in this model comes only from such tasks.

The model does not determine the share of income, which is a separate question. Under the premise, a human service that perfectly substitutes for a free cognitive service cannot command a positive competitive price for its cognitive content, so the share of tasks held by people is an upper bound on the share of income earned from them, and the bound is attained only for tasks whose requirement is not cognitive. Acemoglu and Restrepo model new tasks in which labour has a comparative advantage and state that their framework also includes a regime in which all tasks are automated (Acemoglu and Restrepo 2018), and Autor's account of why many jobs persist is an argument about complementarity under stated conditions (Autor 2015). Neither is at fault. The error arises when a conditional result is repeated without its conditions.

Human fallback after automated incident response

Automated incident response now handles routine failures in production software, and a common concern follows: responders lose the practice that routine incidents provided and are less prepared for the difficult incidents automation cannot handle.

The concern is well founded. Bainbridge's analysis of automation considers systems in which a person must take over under abnormal conditions and notes that automation removes the routine practice that keeps that person capable (Bainbridge 1983). The out-of-the-loop performance problem and its relation to the level of control in automation have been studied directly (Endsley and Kiris 1995). Where a person is a necessary recovery component, reducing that person's readiness reduces the reliability of the system.

The further claim that a person must remain the recovery component does not follow. When the primary system fails, the relevant comparison is between conditional success probabilities, and the value of a fallback depends on how many of the primary system's failures it shares. We model a practised responder with unconditional success 0.72 who also fails on a tenth of the primary's failures, a second automated system with capability κ that shares a fraction ρ of the primary's failures, and a safe-state procedure that contains a failure without diagnosing it and succeeds 0.55 of the time.

Across the plane of capability and shared failure, the person is the best fallback in 84.85 percent of cells, so the intuition behind the concern is usually correct. A second automated system is better wherever its failures are sufficiently independent of the primary's: to outperform the person at any capability up to one, it must share at most 0.352 of the primary's failures. The requirement for a fallback is therefore a requirement for failure independence, which a person often but not always provides. Turing described the general form of this argument in 1950: a demonstrated capability is conceded while some further activity is reserved for people (Turing 1950).

The second result supports Bainbridge's own recommendation. As routine incidents are automated, the responder's conditional success falls linearly, and it drops below the safe-state pro-

cedure at 43.2 percent automation of routine incidents. Beyond that point, in this model, an out-of-practice person does worse than placing the system in a contained state without diagnosis. Bainbridge recommends automatic shutdown where appropriate, and this result gives the corresponding threshold. Loss of skill is therefore a reason to decide explicitly whether a person should remain in the loop.

The third result concerns evidence. If the mean time to resolve incidents handled by people rises after automation, that increase is expected even with competence unchanged, because automation removes the easy cases and leaves a harder sample. With incident difficulty drawn from a gamma distribution and resolution time a linear function of difficulty, automating 75 percent of routine incidents raises mean human resolution time by a factor of 1.88 with no change in competence. A cohort whose competence declines on an unchanged mix of cases shows a factor of 1.36. When both effects are combined, selection accounts for 71 percent of the increase. A rise in resolution time is therefore not evidence of lost skill until the mix of cases is controlled for, since both explanations predict it.

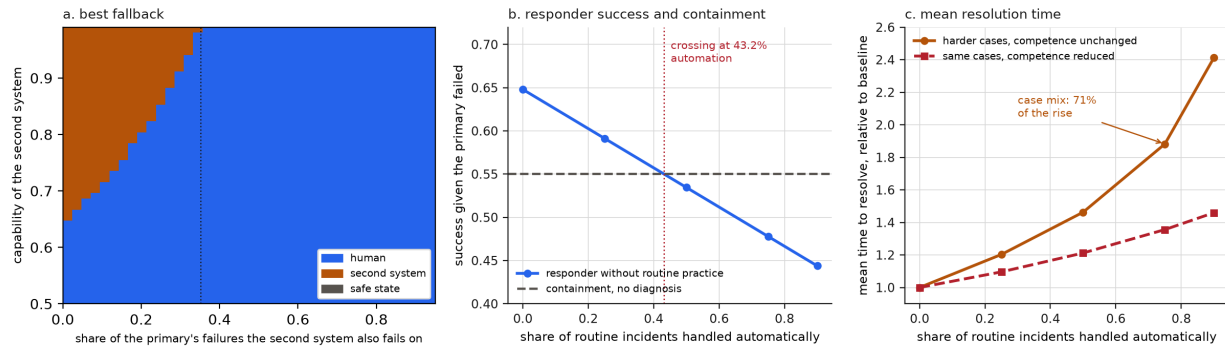


Figure 2: Fallback after automated incident response. (a) The best fallback across machine capability and the share of the primary system's failures that a second system shares; the safe-state procedure is not the best option anywhere in this plane. (b) A responder's conditional success as routine practice is withdrawn, compared with containment. (c) Mean human resolution time under two hypotheses: harder remaining cases with competence unchanged, and an unchanged mix of cases with competence reduced.

Constraints that free cognition does not remove

Some constraints survive the counterfactual. Some impossibility results are not limits of reasoning. Myerson and Satterthwaite showed that under independent private valuations with overlapping supports, no mechanism for bilateral trade is simultaneously incentive compatible, individually rational, budget balanced without subsidy and ex post efficient (Myerson and Satterthwaite 1983). Additional computation does not change this; changing access to information or allowing a subsidy does. Each institutional friction therefore needs its own analysis.

Some mechanisms fail at zero cost instead of degrading smoothly. In a ratio contest for a fixed prize with effort cost ε per unit, the symmetric equilibrium has aggregate expenditure independent of ε

(Tullock 2001). At $\varepsilon = 0$ there is no finite-effort equilibrium, since any participant can improve its odds by deploying more effort without limit, and the allocation rule has to be replaced. Counting agents does not solve the problem, because many identities controlled by one principal are not many claimants, which is the false-name problem in a new setting (Yokoo et al. 2004).

Objectives remain a political matter. Free analysis can compare means to a given end but does not choose the end. Sen distinguished the instrumental role of democracy from its constructive role, in which public deliberation forms the objectives a society then pursues (Sen 1999). Ostrom's polycentric governance (Ostrom 2010) becomes easier to operate under the premise, because the cognitive cost of running many overlapping institutions falls to zero with everything else. An accurate forecast does not confer legitimacy on a constitution.

A single rule applies throughout: whenever an analysis introduces a bottleneck, it should ask whether the bottleneck consists of cognitive work that the premise has made free. If it does, the analysis must explain why the bottleneck survives; if it does not, the analysis should identify which of the four exclusions it belongs to. The rule prevents an argument from reintroducing costly intelligence wherever the existing arrangement requires it, and the models above are subject to it as well.

Implications and limitations

The models are benchmarks and do not forecast. The allocation model holds supplies fixed and has no investment. The task model has no prices and therefore bounds an income share without computing one; its units are tasks. The fallback model prices a single class of incidents with stated success probabilities and no correlated cascades. None of the numbers estimates a real economy.

Prior work anticipated machine-dominated production and stated its premises clearly, and frameworks that include a full-automation regime say so (Acemoglu and Restrepo 2018). The target here is an inference that persists after its conditions are no longer stated. The permanent-remainder assumption is rarely asserted explicitly, but it is often relied on.

The results give concrete tests for the argument. A claim that people will retain a share of output should state whether it rests on a frontier that is never empty, which is compatible with a vanishing share, or on tasks with a non-cognitive requirement, which alone produce a positive limit. A claim that new work will appear should recognise that the volume of new work does not enter the limit. A claim that a person must remain the fallback should state the degree of failure independence that would make it true. A claim that rising resolution times demonstrate lost skill should first control for the mix of cases.

The allocation result has a further implication. With identical intelligence and identical technology, the distribution of consumption ranged from 81 to 1 to 1 to 1 depending only on enforceable claims to assets. What people can do did not determine what they received. Arguments that ground people's economic standing in a continuing ability to outperform machines at some task make that standing depend on a comparison the models show to be unnecessary: distribution is set by ownership and pooling, and these are political choices.

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